Lab Sheet 2

Experiment Name : Exploratory Data Analysis (EDA) on a Real-World Business Dataset using Python

Steps

1.
Select a real-world dataset such as the Netflix Titles Dataset, Superstore Sales Dataset, or IBM HR Employee Attrition Dataset.

2.
Import the required Python libraries (Pandas, NumPy, Matplotlib, and Seaborn) and load the cleaned dataset.

3.
Explore the dataset by examining its dimensions, data types, column names, and summary statistics using functions such as info(), describe(), and value_counts().

4.
Analyze the distribution of numerical features using histograms and density plots to understand central tendency and data spread.

5.
Visualize categorical variables using bar charts and count plots to identify the most frequent categories.

6.
Generate a correlation matrix and heatmap to identify relationships between numerical variables and detect highly correlated features.

7.
Create box plots and scatter plots to identify outliers and study relationships between different variables.

8.
Perform univariate, bivariate, and multivariate analysis to uncover trends and business patterns within the dataset.

9.
Summarize key observations and identify factors that significantly influence the target variable or business objective.

10.
Prepare a concise report highlighting important findings and recommendations based on the exploratory analysis.

Solution ->

STEP 1 and 2 : Import Libraries and Load Dataset

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
url = "https://raw.githubusercontent.com/IBM/employee-attrition-aif360/master/data/emp_attrition.csv"
df = pd.read_csv(url)
print("Dataset loaded successfully!")
print("Shape:", df.shape)
display(df.head())
```

```
Dataset loaded successfully!
Shape: (1470, 35)
```

	Age	Attrition	BusinessTravel	DailyRate	Department	DistanceFromHome	Education	EducationField	EmployeeCount	EmployeeNumber	...	RelationshipSatisfaction	StandardHours	StockOptionLevel	TotalWorkingYears	Traini
0	41	Yes	Travel_Rarely	1102	Sales	1	2	Life Sciences	1	1	...	1	80	0	8	
1	49	No	Travel_Frequently	279	Research & Development	8	1	Life Sciences	1	2	...	4	80	1	10	
2	37	Yes	Travel_Rarely	1373	Research & Development	2	2	Other	1	4	...	2	80	0	7	
3	33	No	Travel_Frequently	1392	Research & Development	3	4	Life Sciences	1	5	...	3	80	0	8	
4	27	No	Travel_Rarely	591	Research & Development	2	1	Medical	1	7	...	4	80	1	6	

5 rows × 35 columns

STEP 3 : Explore the Dataset

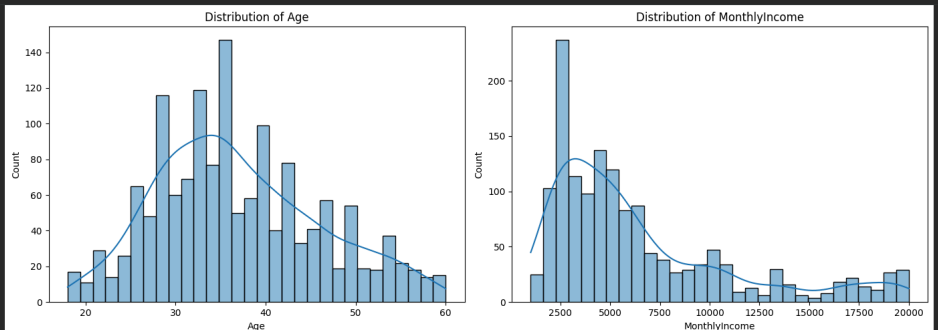
```
print("Number of Rows:", df.shape[0])
print("Number of Columns:", df.shape[1])
print("Column Names:")
print(df.columns.tolist())
print("Dataset Information:")
df.info()
print("Summary Statistics:")
display(df.describe())
print("Attrition Distribution:")
print(df["Attrition"].value_counts())
print("\nDepartment Distribution:")
print(df["Department"].value_counts())
print("\nGender Distribution:")
print(df["Gender"].value_counts())
print("\nJob Role Distribution:")
print(df["JobRole"].value_counts())
```

```
 31  YearsAtCompany            1470 non-null   int64
 32  YearsInCurrentRole        1470 non-null   int64
 33  YearsSinceLastPromotion   1470 non-null   int64
 34  YearsWithCurrManager      1470 non-null   int64
dtypes: int64(26), object(9)
memory usage: 402.1+ KB
Summary Statistics:
```

	Age	DailyRate	DistanceFromHome	Education	EmployeeCount	EmployeeNumber	EnvironmentSatisfaction	HourlyRate	JobInvolvement	JobLevel	...	RelationshipSatisfaction	StandardHours	StockOptionLevel
count	1470.000000	1470.000000	1470.000000	1470.000000	1470.0	1470.000000	1470.000000	1470.000000	1470.000000	1470.000000	...	1470.000000	1470.0	1470.000000
mean	36.923810	802.485714	9.192517	2.912925	1.0	1024.865306	2.721769	65.891156	2.729932	2.063946	...	2.712245	80.0	0.793878
std	9.135373	403.509100	8.106864	1.024165	0.0	602.024335	1.093082	20.329428	0.711561	1.106940	...	1.081209	0.0	0.852077
min	18.000000	102.000000	1.000000	1.000000	1.0	1.000000	1.000000	30.000000	1.000000	1.000000	...	1.000000	80.0	0.000000
25%	30.000000	465.000000	2.000000	2.000000	1.0	491.250000	2.000000	48.000000	2.000000	1.000000	...	2.000000	80.0	0.000000

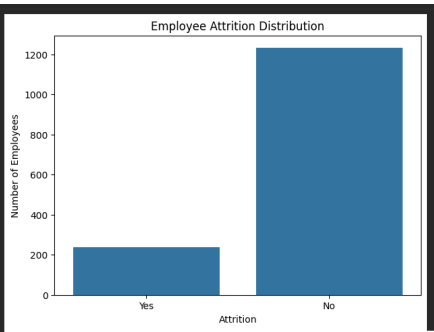
STEP 4 : Analyze Numerical Features Using Histograms and Density Plots

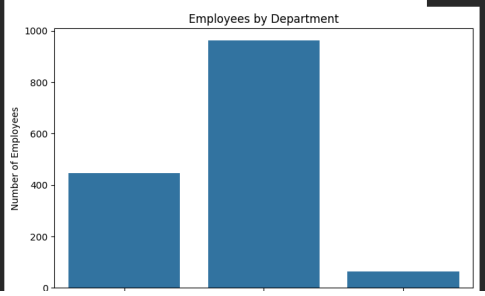
```
numerical_features = [
    "Age",
    "MonthlyIncome",
    "TotalWorkingYears",
    "YearsAtCompany"
]
plt.figure(figsize=(14, 10))
for i, column in enumerate(numerical_features, 1):
    plt.subplot(2, 2, i)
    sns.histplot(
        df[column],
        kde=True,
        bins=30
    )
    plt.title(f"Distribution of {column}")
plt.tight_layout()
plt.show()
```



STEP 5 : Visualize Categorical Variables

```
plt.figure(figsize=(7, 5))
sns.countplot(
    data=df,
    x="Attrition"
)
plt.title("Employee Attrition Distribution")
plt.xlabel("Attrition")
plt.ylabel("Number of Employees")
plt.show()
plt.figure(figsize=(8, 5))
sns.countplot(
    data=df,
    x="Department"
)
plt.title("Employees by Department")
plt.xlabel("Department")
plt.ylabel("Number of Employees")
plt.xticks(rotation=20)
plt.show()
plt.figure(figsize=(12, 6))
sns.countplot(
    data=df,
    y="JobRole",
    order=df["JobRole"].value_counts().index
)
plt.title("Employees by Job Role")
plt.xlabel("Number of Employees")
plt.ylabel("Job Role")
plt.show()
plt.figure(figsize=(7, 5))
sns.countplot(
    data=df,
    x="Gender"
)
plt.title("Gender Distribution")
plt.xlabel("Gender")
plt.ylabel("Number of Employees")
plt.show()
```

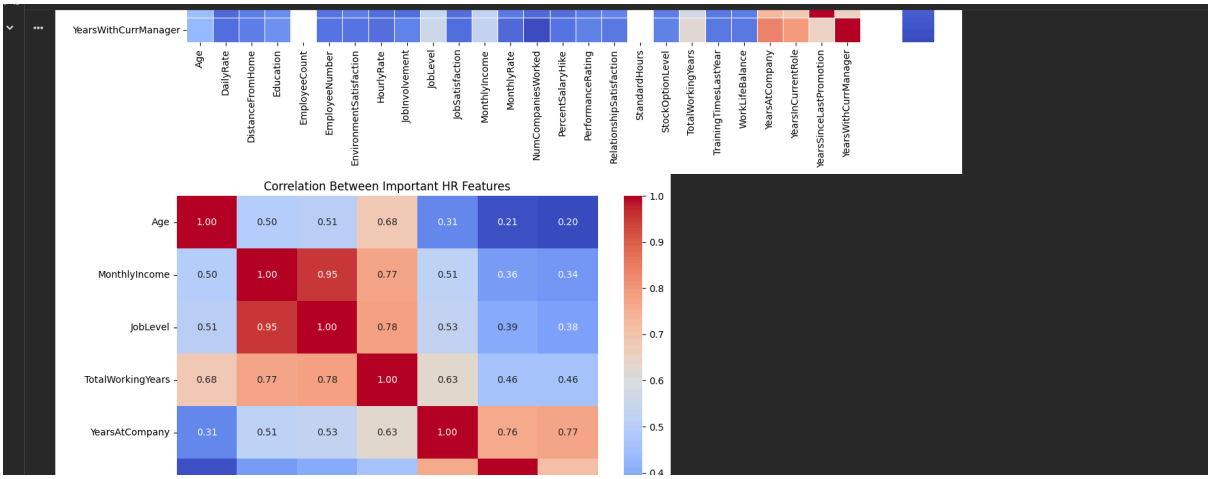




STEP 6 : Correlation Matrix and Heatmap

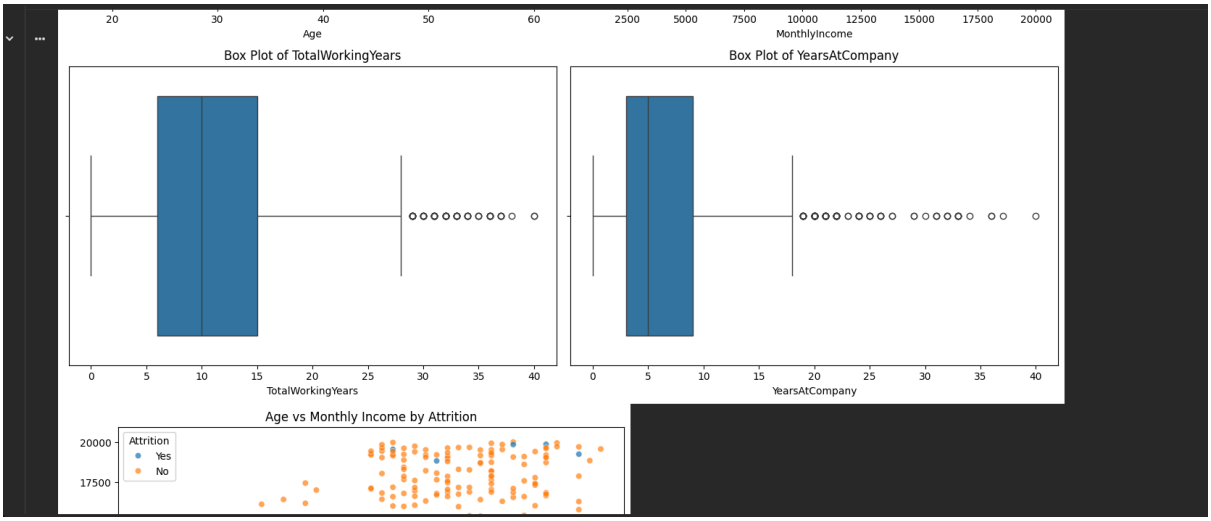
```
[6]
✓ 1s
```

```python
numerical_df = df.select_dtypes(
    include=np.number
)
print("Numerical Columns:")
print(numerical_df.columns.tolist())
correlation_matrix = numerical_df.corr()
display(correlation_matrix)
plt.figure(figsize=(16, 12))
sns.heatmap(
    correlation_matrix,
    cmap="coolwarm",
    annot=False,
    linewidths=0.5
)
plt.title("Correlation Heatmap of Numerical Features")
plt.show()
important_features = [
    "Age",
    "MonthlyIncome",
    "JobLevel",
    "TotalWorkingYears",
    "YearsAtCompany",
    "YearsInCurrentRole",
    "YearsWithCurrManager"
]
plt.figure(figsize=(10, 7))
sns.heatmap(
    df[important_features].corr(),
    annot=True,
    cmap="coolwarm",
    fmt=".2f"
)
plt.title("Correlation Between Important HR Features")
plt.show()
```



STEP 7 : Box Plots and Scatter Plots

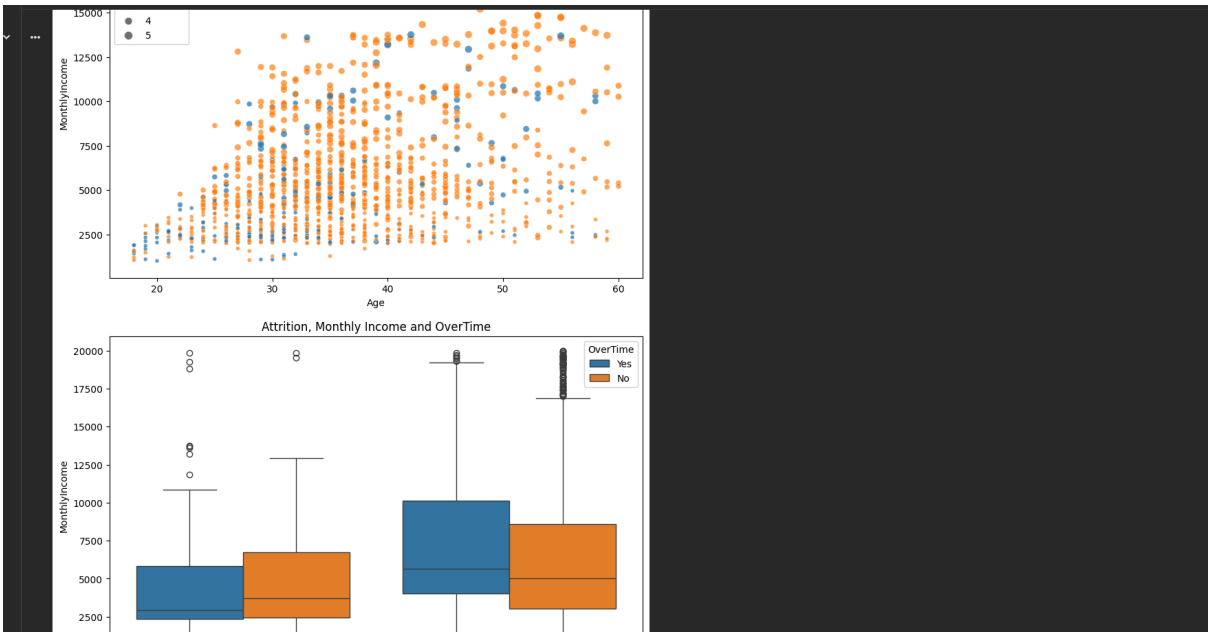
```
outlier_features = [
    "Age",
    "MonthlyIncome",
    "TotalWorkingYears",
    "YearsAtCompany"
]
plt.figure(figsize=(14, 10))
for i, column in enumerate(outlier_features, 1):
    plt.subplot(2, 2, i)
    sns.boxplot(
        x=df[column]
    )
    plt.title(f"Box Plot of {column}")
plt.tight_layout()
plt.show()
plt.figure(figsize=(9, 6))
sns.scatterplot(
    data=df,
    x="Age",
    y="MonthlyIncome",
    hue="Attrition",
    alpha=0.7
)
plt.title("Age vs Monthly Income by Attrition")
plt.xlabel("Age")
plt.ylabel("Monthly Income")
plt.show()
plt.figure(figsize=(9, 6))
sns.scatterplot(
    data=df,
    x="TotalWorkingYears",
    y="MonthlyIncome",
    hue="Attrition",
    alpha=0.7
)
plt.title("Total Working Years vs Monthly Income by Attrition")
plt.xlabel("Total Working Years")
plt.ylabel("Monthly Income")
plt.show()
```



STEP 8 : Univariate, Bivariate and Multivariate Analysis

```
print("Attrition Counts:")
print(df["Attrition"].value_counts())
plt.figure(figsize=(7, 5))
df["Attrition"].value_counts().plot(
    kind="bar"
)
plt.title("Attrition Distribution")
plt.xlabel("Attrition")
plt.ylabel("Number of Employees")
plt.xticks(rotation=0)
plt.show()
plt.figure(figsize=(8, 6))
sns.boxplot(
    data=df,
    x="Attrition",
    y="MonthlyIncome"
)
plt.title("Monthly Income vs Attrition")
plt.show()
plt.figure(figsize=(8, 6))
sns.boxplot(
    data=df,
    x="Attrition",
    y="Age"
)
plt.title("Age vs Attrition")
plt.show()
plt.figure(figsize=(8, 5))
sns.countplot(
    data=df,
    x="OverTime",
    hue="Attrition"
)
plt.title("OverTime vs Attrition")
plt.xlabel("OverTime")
plt.ylabel("Number of Employees")
plt.show()
plt.figure(figsize=(9, 6))
sns.countplot(
    data=df,
    x="Department",
    hue="Attrition"
)
```

```
plt.title("Age vs Attrition")
plt.show()
plt.figure(figsize=(8, 5))
sns.countplot(
    data=df,
    x="OverTime",
    hue="Attrition"
)
plt.title("OverTime vs Attrition")
plt.xlabel("OverTime")
plt.ylabel("Number of Employees")
plt.show()
plt.figure(figsize=(9, 6))
sns.countplot(
    data=df,
    x="Department",
    hue="Attrition"
)
plt.title("Department vs Attrition")
plt.xticks(rotation=20)
plt.show()
plt.figure(figsize=(10, 7))
sns.scatterplot(
    data=df,
    x="Age",
    y="MonthlyIncome",
    hue="Attrition",
    size="JobLevel",
    alpha=0.7
)
plt.title("Age, Income, Job Level and Attrition")
plt.show()
plt.figure(figsize=(10, 6))
sns.boxplot(
    data=df,
    x="Attrition",
    y="MonthlyIncome",
    hue="OverTime"
)
plt.title("Attrition, Monthly Income and OverTime")
plt.show()
```



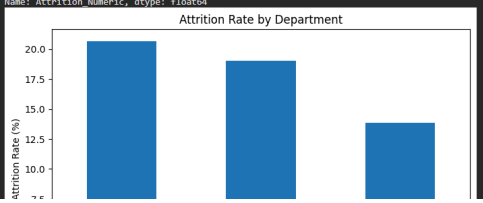
STEP 9 : Identify Factors Influencing Attrition

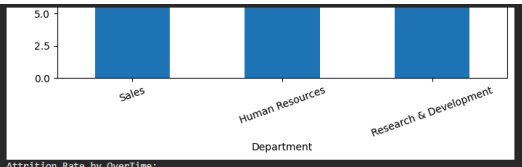
```
df_analysis = df.copy()
df_analysis["Attrition_Numeric"] = df_analysis["Attrition"].map(
    {
        "Yes": 1,
        "No": 0
    }
)
print("Attrition Rate:")
print(
    df_analysis["Attrition_Numeric"].mean() * 100
)
department_attrition = (
    df_analysis.groupby("Department")["Attrition_Numeric"]
    .mean()
    .sort_values(ascending=False)
    * 100
)
print("Attrition Rate by Department:")
print(department_attrition.round(2))
plt.figure(figsize=(8, 5))
department_attrition.plot(
    kind="bar"
)
plt.title("Attrition Rate by Department")
plt.xlabel("Department")
plt.ylabel("Attrition Rate (%)")
plt.xticks(rotation=20)
plt.show()
overtime_attrition = (
    df_analysis.groupby("OverTime")["Attrition_Numeric"]
    .mean()
    * 100
)
print("Attrition Rate by OverTime:")
print(overtime_attrition.round(2))
plt.figure(figsize=(7, 5))

overtime_attrition.plot(
    kind="bar"
)
plt.title("Attrition Rate by OverTime")
plt.xlabel("OverTime")
plt.ylabel("Attrition Rate (%)")
plt.xticks(rotation=0)
```

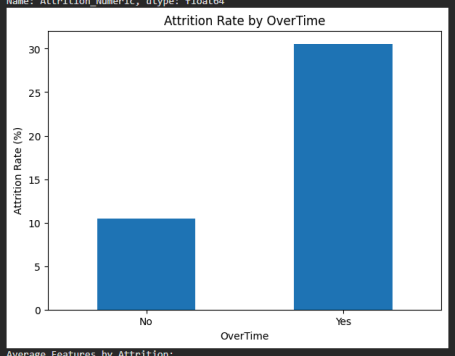
```
overtime_attrition.plot(
    kind="bar"
)
plt.title("Attrition Rate by OverTime")
plt.xlabel("OverTime")
plt.ylabel("Attrition Rate (%)")
plt.xticks(rotation=0)
plt.show()
attrition_comparison = (
    df_analysis.groupby("Attrition")[
        [
            "Age",
            "MonthlyIncome",
            "JobLevel",
            "TotalWorkingYears",
            "YearsAtCompany",
            "JobSatisfaction",
            "WorkLifeBalance"
        ]
    ]
    .mean()
)
print("Average Features by Attrition:")
display(attrition_comparison)
```

```
Attrition Rate:
16.12244897959183
Attrition Rate by Department:
Department
Sales                     20.63
Human Resources           19.05
Research & Development    13.84
Name: Attrition_Numeric, dtype: float64
```






```
Attrition Rate by OverTime:
OverTime
No     10.44
Yes    30.53
Name: Attrition_Numeric, dtype: float64
```

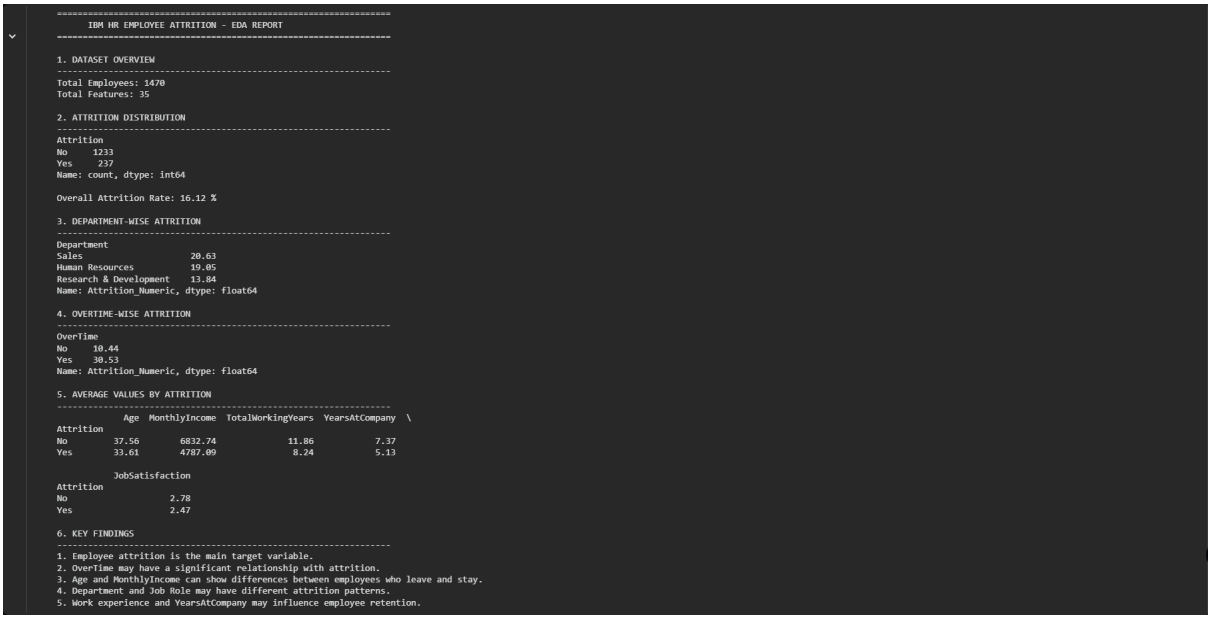


Average Features by Attrition:

Attrition	Age	MonthlyIncome	JobLevel	TotalWorkingYears	YearsAtCompany	JobSatisfaction	WorkLifeBalance
No	37.561233	6832.739659	2.145985	11.862936	7.369019	2.778589	2.781022
Yes	33.607595	4787.092827	1.637131	8.244726	5.130802	2.468354	2.658228

STEP 10 : Concise report

```
print("=" * 65)
print("      IBM HR EMPLOYEE ATTRITION - EDA REPORT")
print("=" * 65)
print("\n1. DATASET OVERVIEW")
print("-" * 65)
print("Total Employees:", len(df))
print("Total Features:", len(df.columns))
print("\n2. ATTRITION DISTRIBUTION")
print("-" * 65)
print(df["Attrition"].value_counts())
attrition_rate = (
    df["Attrition"]
    .value_counts(normalize=True)["Yes"]
    * 100
)
print("\nOverall Attrition Rate:", round(attrition_rate, 2), "%")
print("\n3. DEPARTMENT-WISE ATTRITION")
print("-" * 65)
print(department_attrition.round(2))
print("\n4. OVERTIME-WISE ATTRITION")
print("-" * 65)
print(overtime_attrition.round(2))
print("\n5. AVERAGE VALUES BY ATTRITION")
print("-" * 65)
print(
    attrition_comparison[
        [
            "Age",
            "MonthlyIncome",
            "TotalWorkingYears",
            "YearsAtCompany",
            "JobSatisfaction"
        ]
    ].round(2)
)
print("\n6. KEY FINDINGS")
print("-" * 65)
print("1. Employee attrition is the main target variable.")
print("2. OverTime may have a significant relationship with attrition.")
print("3. Age and MonthlyIncome can show differences between employees who leave and stay.")
print("4. Department and Job Role may have different attrition patterns.")
print("5. Work experience and YearsAtCompany may influence employee retention.")
print("\n7. RECOMMENDATIONS")
print("-" * 65)
print("1. Monitor employees who frequently work overtime.")
```



```
======================================================================
      IBM HR EMPLOYEE ATTRITION - EDA REPORT
======================================================================

1. DATASET OVERVIEW
----------------------------------------------------------------------
Total Employees: 1470
Total Features: 35

2. ATTRITION DISTRIBUTION
----------------------------------------------------------------------
Attrition
No     1233
Yes     237
Name: count, dtype: int64

Overall Attrition Rate: 16.12 %

3. DEPARTMENT-WISE ATTRITION
----------------------------------------------------------------------
Department
Sales                  20.63
Human Resources        19.05
Research & Development    13.84
Name: Attrition_Numeric, dtype: float64

4. OVERTIME-WISE ATTRITION
----------------------------------------------------------------------
OverTime
No     10.44
Yes    30.53
Name: Attrition_Numeric, dtype: float64

5. AVERAGE VALUES BY ATTRITION
----------------------------------------------------------------------
               Age  MonthlyIncome  TotalWorkingYears  YearsAtCompany  \
Attrition
No         37.56        6832.74              11.86            7.37
Yes        33.61        4787.09               8.24            5.13

           JobSatisfaction
Attrition
No                    2.78
Yes                   2.47

6. KEY FINDINGS
----------------------------------------------------------------------
1. Employee attrition is the main target variable.
2. OverTime may have a significant relationship with attrition.
3. Age and MonthlyIncome can show differences between employees who leave and stay.
4. Department and Job Role may have different attrition patterns.
5. Work experience and YearsAtCompany may influence employee retention.
```

Questions : **1. What is Exploratory Data Analysis (EDA), and why is it performed before machine learning?**

EDA is the process of examining and visualizing data to understand its patterns, relationships, and quality. It is performed before machine learning to **detect missing values, outliers, errors, and important features**.

2. Differentiate between univariate, bivariate, and multivariate analysis.

- **Univariate:** Analysis of one variable. Example: distribution of employee ages.
- **Bivariate:** Analysis of two variables. Example: Age vs. Monthly Income.
- **Multivariate:** Analysis of three or more variables. Example: Age, Income, and Attrition together.

3. What insights can be obtained from a correlation heatmap?

A correlation heatmap shows the **strength and direction of relationships** between numerical variables. It helps identify highly correlated features, possible redundancy, and useful relationships for prediction.

4. Explain the purpose of histograms, box plots, and scatter plots in EDA.

- **Histogram:** Shows the distribution and frequency of numerical data.
- **Box Plot:** Shows median, spread, and potential outliers.
- **Scatter Plot:** Shows the relationship or pattern between two numerical variables.

5. How can EDA help identify data quality issues before analysis?

EDA can reveal **missing values, duplicate records, incorrect data types, inconsistent categories, outliers, and unusual distributions**. These issues can then be corrected before analysis or model building.

6. Why is correlation important in predictive analytics? Can correlation imply causation?

Correlation helps identify variables that may be useful for **prediction and feature selection**. However, **correlation does not imply causation** because two variables can be related due to other factors.

7. Which visualization would you use to analyze categorical and numerical variables? Justify your choice.

- **Categorical:** Bar chart or count plot because they clearly show category frequencies.
- **Numerical:** Histogram or box plot because they show distribution, spread, and outliers.
- **Categorical + numerical:** Box plot is useful for comparing numerical values across categories.

8. What business insights can be derived from the Netflix/HR Analytics dataset through EDA?

From **HR Analytics**, EDA can identify factors related to employee attrition, such as **overtime, job role, income, age, job satisfaction, and years at the company**. These insights can help organizations improve employee retention.

9. How does EDA contribute to feature selection and model building?

EDA helps identify **relevant features, highly correlated variables, outliers, and data patterns**. Irrelevant or redundant features can be removed, resulting in simpler and potentially more accurate machine learning models.

10. What challenges might arise while performing EDA on large-scale real-world datasets?

Major challenges include **large data size, slow processing, missing or inconsistent data, high-dimensional features, memory limitations, and difficulty in visualizing huge datasets**. Efficient tools and sampling techniques may be required.